ENGR 1451/2451: Exploratory Data Science

Simple Linear Regression
Slope Inference and Prediction

Prof. Paul W. Leu
University of Pittsburgh

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Today: From a Fitted Line to an Inference

Brief review of simple linear regression

Residuals, R^2, and diagnostic plots tell us what a fitted line describes—and what it misses.

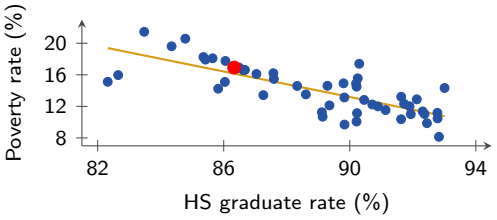
New question

A sample has a nonzero slope. Is that convincing evidence that the *population* slope is nonzero?

What we will learn

Check residuals, make inference for the slope, then compare **confidence intervals for the mean** and **prediction intervals for individuals**.

Review: What Did the State Regression Tell Us?



Fitted line

$$\widehat{y} \approx 85.789 - 0.807x$$

One residual

$e_i = y_i - \widehat{y}_i$

Georgia: $e_i \approx +0.77$ pp.

Two fit summaries

RMSE (typical residual size):

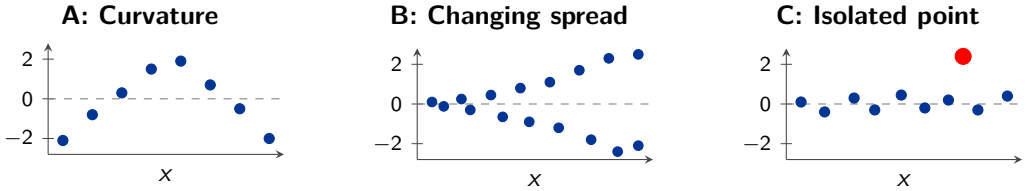
$$\mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n} e_i^2} \approx 1.90 \text{ pp}$$

$R^2 \approx 0.599$**:**

About 60% of the observed poverty-rate variation is accounted for by the line.

Instructor (25 sec): RMSE is in poverty percentage points; R^2 is unitless. This association is not causal. Later, regression inference uses residual SD with $n - 2$ instead of n.

Review: What Do Residual Plots Diagnose?



Which plot questions the **straight-line mean**, which questions **constant variance**, and which calls for investigation of an **unusual observation**?

Instructor (1–2 min): A: linear mean; B: equal spread; C: unusual point. A large residual is not automatically an influential point.

Describing a Sample Versus Inference

What we already know

$$\hat{y} = b_0 + b_1 x$$

b_1: **sample slope** computed from our observed data.

What we want to learn

$$E(Y \mid x) = \beta_0 + \beta_1 x$$

β_1: **population slope**, which we do not know.

A fitted line always exists

Even data with a weak or misleading pattern can produce a nonzero b_1.

Inference asks

Could sampling variation explain the observed b_1, if the true slope β_1 were zero?

Instructor (30 sec): The parameter β_1 is fixed but unknown; its estimate b_1 varies from sample to sample.

Regression Model Assumptions

A model for the response at a fixed predictor value

$$Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i, \qquad \varepsilon_i \overset{\text{ind}}{\sim} N(0, \sigma^2).$$

1. Straight-line mean
$E(Y_i \mid x_i) = \beta_0 + \beta_1 x_i$.

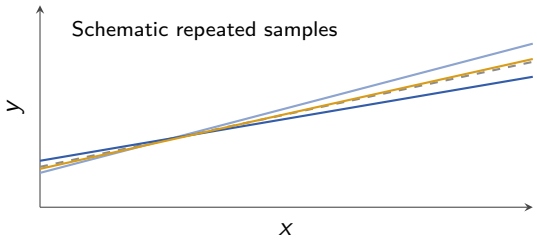
2. Independent observations
The errors for distinct observations are independent.

3. Roughly constant spread
Error SD σ does not depend on x.

4. Nearly Gaussian errors
Especially helpful for small samples.

Instructor (45 sec): Normality concerns the errors at a fixed x, not the marginal distribution of x or y. Residual plots from w06b help assess these model conditions.

What Is the Standard Error of a Slope?



Imagine repeating the study

Each random sample can give a slightly different fitted slope b_1.

Standard error

$SE(b_1)$ estimates the sample-to-sample spread of those slopes.

Key idea: A large $|b_1|$ matters more when $SE(b_1)$ is small.

Example: 27 Pairs of Identical Twins

Historical textbook dataset

x: IQ of the twin raised by biological parents.

y: IQ of the twin raised by foster parents.

$n = 27$ twin pairs; sample correlation $r = 0.882$.

Fitted model

$$\widehat{\text{foster IQ}} = 9.2076 + 0.9014x$$

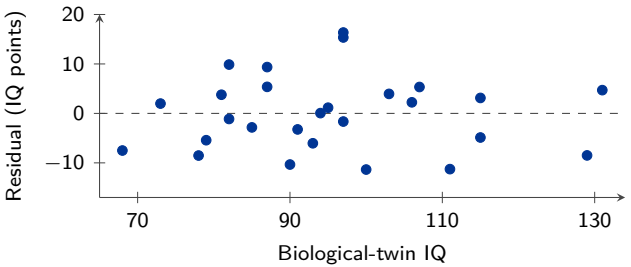
Questions

1. Is the population slope different from 0?
2. What slopes are compatible with the data?
3. Can we generalize beyond these 27 pairs?

$R^2 = 0.7779$ describes **in-sample fit**, not a causal effect.

Instructor (30 sec): OpenIntro treats this as an illustrative example under an assumed representative sample. Historical IQ data are not proof of genetic causation; inference and causation are different questions.

Check the Twin Residuals Before Inference



Observed minus fitted:

$e_i = y_i - \widehat{y}_i$.

Look for:

- Curvature.
- Changing vertical spread.
- Isolated observations.

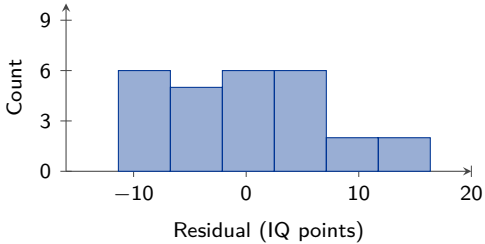
Discuss: Which concerns do you see here?

```
plt.scatter(twins["biological"], model.resid)
plt.axhline(0, linestyle="--")
plt.show()
```

Instructor (40 sec): No strong curve or clear fan is apparent, but the sample is small. Several residuals deserve attention. A roughly patternless plot does not prove the assumptions.

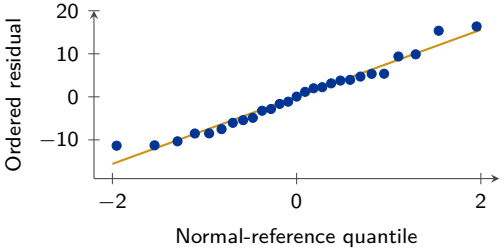
Check the Shape of the Residuals

Histogram



Look for asymmetry and extreme residuals.

Normal probability plot



Roughly straight points support a normal error shape.

Check residuals, not the raw x or y values. Neither plot establishes independence or representative sampling.

Instructor (40 sec): The points are broadly near the reference line, with some tail departures; the histogram has some right asymmetry. With 27 pairs, treat these as visual checks, not proof of normality.

Sensitivity Check: Does One Pair Drive the Slope?

Refit without the observation with the largest absolute residual. Here it is the pair $(x, y) = (97, 113)$, with residual about $+16.35$ IQ points.

```
pair_to_check = model.resid.abs().idxmax()
model_without = smf.ols(
    "foster ~ biological", data=twins.drop(index=pair_to_check)
).fit()
```

Data used	n	Intercept	Slope
All pairs	27	9.2076	0.9014
Without selected pair	26	9.0068	0.8969

Does this pair materially change the fitted slope?

Instructor (35 sec): The slope changes only from about 0.901 to 0.897. This particular pair does not drive the slope. The check does not assess every potentially influential observation.

Discussion: Which Hypotheses Test the Slope?

We want to know whether the **population mean response** has a linear relationship with x.

Choose one pair of hypotheses

1. $H_0 : b_0 = 0, \quad H_A : b_0 \neq 0$
2. $H_0 : \beta_0 = 0, \quad H_A : \beta_0 \neq 0$
3. $H_0 : b_1 = 0, \quad H_A : b_1 \neq 0$
4. $H_0 : \beta_1 = 0, \quad H_A : \beta_1 \neq 0$

Instructor (1–2 min): (4). β_1 is the unknown population slope; b_1 is the observed sample statistic. Zero slope means no *linear* mean trend, not necessarily no other relationship.

A t-Test for the Population Slope

Test statistic

$$T = \frac{b_1 - \beta_{1,0}}{SE(b_1)}, \qquad \boxed{df = n - 2}$$

The slope value specified by the null hypothesis, usually zero. **For a no-trend null:** $\beta_{1,0} = 0$.

A two-sided test compares $|T|$ to a t distribution with $n - 2$ degrees of freedom.

Why $n - 2$?

We fit **two coefficients**:

β_0 (intercept)

β_1 (slope).

For $n = 27$ twin pairs:

$df = 27 - 2 = 25$.

Instructor (30 sec): The test asks how many slope standard errors the estimated slope is from the null value. Conventional simple-OLS inference uses this t test.

Read the Regression Table: Which Row Matters?

Model for 27 twin pairs

	Estimate	Std. Error	t value	Two-sided p
Intercept	9.2076	9.2999	0.99	0.3316
Biological twin IQ	**0.9014**	**0.0963**	**9.36**	$\mathbf{1.2 \times 10^{-9}}$

Estimate $b_1 = 0.9014$
One additional biological-twin IQ point is *associated* with about 0.90 additional fitted foster-twin IQ points.

Standard error $SE(b_1) = 0.0963$
How uncertain is the slope estimate under the sampling/model assumptions?

Instructor (30 sec): We generally focus on the slope row. The intercept's p-value tests a different question: whether mean y at $x = 0$ is zero.

Discussion: What Does R^2 Really Mean?

The twin regression has $R^2 = 0.7779$ and $\widehat{y} = 9.2076 + 0.9014x$, where x is biological-twin IQ.

Which statement is *false*?

1. A 10-point increase in x corresponds to about 9 additional fitted foster-IQ points.
2. The line accurately predicts approximately 78% of *individual* foster-IQ scores.
3. The model can be written as $\widehat{y} \approx 9.2 + 0.90x$.
4. About 78% of the *observed variation* in foster IQ is accounted for by the fitted line.

Instructor (1–2 min): (2) is false. R^2 measures variation accounted for in the observed sample; it is not an accuracy rate, the probability a prediction is correct, or a causal claim.

Question adapted from OpenIntro Statistics (2nd ed.), Ch. 7 regression-output discussion (CC BY-SA).

Can We Test the y-Intercept Too?

Yes: use the same t logic

$$H_0 : \beta_0 = 0, \qquad H_A : \beta_0 \neq 0$$

$$T = \frac{b_0}{SE(b_0)}, \qquad df = n - 2$$

Twin regression output

$b_0 = 9.2076$, $SE(b_0) = 9.2999$

$t \approx 0.99$, $p \approx 0.332$.

But what does it ask?

Is the *mean foster IQ* at biological-twin IQ $x = 0$ different from zero?

The scientific question matters

Here the observed x range is 68–131, so $x = 0$ is extreme extrapolation.

In **sensor calibration**, testing a zero-concentration signal can be useful.

Instructor (35 sec): Both slope and intercept have standard errors and confidence intervals. We focus on the slope because it more often answers the substantive question.

Work the Slope Test

Under $H_0 : \beta_1 = 0$

$$T = \frac{0.9014 - 0}{0.0963} = 9.36$$

$$df = 27 - 2 = 25$$

Two-sided p-value

$$p = P(|T_{25}| \geq 9.36)$$

$$\boxed{p \approx 1.2 \times 10^{-9}}$$

Conclusion—conditional on assumptions

The data provide extremely strong evidence against a zero *linear* population slope.

Statistical significance does **not** establish:

- Causality.
- Accurate predictions for individual twin pairs.

Discussion: Interpret a Very Small p-Value

The slope test reports $p \approx 1.2 \times 10^{-9}$.

Which interpretation is correct?

1. The probability that the population slope equals zero is about 1.2×10^{-9}.
2. If the true population slope were zero and the model held, a result at least this extreme would be extraordinarily unlikely.
3. The slope is so significant that biological IQ must causally determine foster IQ.

Instructor (1–2 min): (2). A p-value conditions on H_0 and a valid model; it is not $P(H_0 \mid \text{data})$. Neither statistical significance nor the twin pairing identifies causality.

A Confidence Interval for the Slope

General 95% confidence interval

$$b_1 \pm t^*_{df=n-2}\, SE(b_1)$$

For 27 observations:

$df = 25$, and $t^*_{25} \approx 2.06$ for 95% confidence.

From the regression output

$$b_1 = 0.9014,$$
$$SE(b_1) = 0.0963.$$

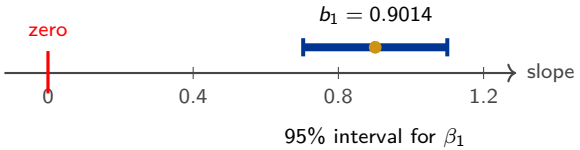
Margin of error

$2.06(0.0963) \approx 0.198.$

Instructor (30 sec): Use a t critical value, not automatically 1.96, because the residual variance is estimated; the degrees of freedom are $n - 2$.

Calculate and Interpret the 95% Interval

$$0.9014 \pm 2.06(0.0963) \approx \boxed{(0.703,\ 1.100)}$$



In context

Under the assumptions, plausible population slopes are about 0.70 to 1.10 foster-IQ points per biological-IQ point.

Connection: The interval excludes 0, agreeing with rejection of $H_0 : \beta_1 = 0$ by a two-sided 5% test.

Instructor (30 sec): A 95% interval refers to the long-run coverage of this procedure; it does not mean 95% of individual data points fall in this interval.

Three Different Intervals in Regression

1. Confidence interval for the slope β_1

A *single range of plausible population slopes*: 0.703 to 1.100 IQ points per IQ point for our example.

2. Confidence interval for the mean at x_0

How uncertain is the **population average** $E(Y \mid x_0)$?

Plot many such intervals to form a **confidence band** around the line.

3. Prediction interval at x_0

Where might **one new response** Y_{new} fall?

This band is **wider**: individuals vary around the mean line.

Instructor (35 sec): A confidence interval for the slope is NOT the narrow confidence band drawn around the fitted line. These estimate different quantities.

Confidence Interval vs. Prediction Interval

At a chosen predictor value x_0, the fitted line gives

$$\boxed{\widehat{y}_0 = b_0 + b_1 x_0}.$$

Confidence interval: population mean response

What is the **average response** when $x = x_0$?

This measures uncertainty in the estimated height of the mean line.

Prediction interval: one new response

Where might **one new observation** fall when $x = x_0$?

This also includes individual scatter around the mean line.

Same center, different uncertainty: the prediction interval is wider.

Instructor (30 sec): The mean-response confidence interval is not an interval for an individual or for the slope.

Understanding the Terms in the Formulas

Symbol	Meaning
n	Number of observations in the sample
x_0	Predictor value of interest
$\bar{x}$	Average predictor value in the sample
$\hat{y}_0$	Predicted response from the fitted line at x_0
$e_i = y_i - \hat{y}_i$	Residual: observed minus predicted response
$s = \sqrt{\frac{\sum_i e_i^2}{n-2}}$	Residual standard deviation: typical vertical scatter around the fitted line
$S_{xx} = \sum_i (x_i - \bar{x})^2$	Total squared spread of the predictor values
t^*	Critical value from the t-distribution; for a 95% interval, $t^* = t_{0.975,n-2}$

Instructor (35 sec): Point out that s measures the scatter of observations around the line. Sxx measures the spread of x-values, not y-values.

Confidence Interval for the Mean Response

How precisely can we estimate the **mean response** at x_0?

95% confidence interval

$$\hat{y}_0 \pm t^* s \sqrt{\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}}$$

What determines the interval width?

- s**:** More scatter around the fitted line makes the interval wider.
- $1/n$**:** More observations generally reduce uncertainty.
- $(x_0 - \bar{x})^2/S_{xx}$**:** Predictions farther from the average predictor value $\bar{x}$ are less precise.

The confidence interval is narrowest at $x_0 = \bar{x}$.

Instructor (35 sec): The interval measures uncertainty in the population mean response. With enough data, we can estimate the mean very precisely, even if individual observations vary substantially.

Prediction Interval for One New Observation

A new observation can fall above or below the line, even when the mean is estimated precisely.

95% prediction interval

$$\hat{y}_0 \pm t^* s \sqrt{1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}}$$

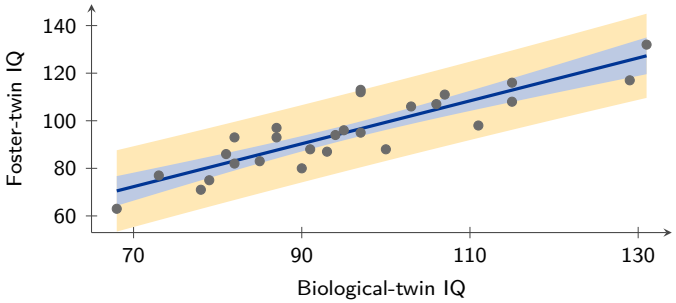
Why the extra 1?

- The mean CI includes uncertainty in the estimated mean.
- The PI also includes **random scatter of one new observation**.
- The extra 1 represents this additional source of variation.

More data can locate the mean more precisely; individual scatter remains.

Instructor (35 sec): Both interval formulas use the independent, constant-variance normal-error model. The additional variance is s^2: factoring out s^2 leaves the extra 1.

See the Difference: Mean CI vs. Individual PI



At $x_0 = 100$

Fitted mean:

$\widehat{y}_0 = 99.35$

Mean 95% CI

[96.15, 102.55]

Individual 95% PI

[83.11, 115.59]

Mean CI

Individual PI

Instructor (30 sec): Read the bands vertically at a given x. The PI targets one new twin; the narrower CI targets the population average. Bands are pointwise 95% intervals, not simultaneous curve coverage. Neither gives causation.

Discussion: Which Interval Answers the Question?

Assume the regression model is reasonable for predicting foster IQ from biological-twin IQ at $x_0 = 100$.

Choose a 95% mean confidence interval or a 95% prediction interval

1. We want to estimate the average foster IQ among people with $x_0 = 100$.
2. We want a range for the foster IQ of one additional twin with $x_0 = 100$.
3. A researcher wants an interval for the population slope, regardless of x_0.

Instructor (1–2 min): (1) Mean-response CI $[96.15, 102.55]$. (2) Individual PI $[83.11, 115.59]$. (3) Neither: use the slope CI $[0.703, 1.100]$. Ask why individual prediction is much less precise.

Small p-Value, But Can We Trust the Inference?

Model conditions

- Residual curvature $\rightarrow$ mean may not be linear.
- Fan shape $\rightarrow$ variance is not constant.
- Extreme residuals $\rightarrow$ assess normality and influence.

Study design

- Are observations independent?
- Are cases randomly sampled or representative?
- What population is being inferred about?

An extremely small computed p-value cannot fix biased selection or a misspecified model.

Instructor (45 sec): For the twin data, the observational units are *pairs*; independence is between pairs, not within each pair. The representativeness assumption was explicitly added in the textbook example.

Discussion: Should We Test the 51 State Slopes?

Return to our states poverty example

Our plotted observations cover the **50 states and DC** (2017 aggregates). We did not randomly sample states from a larger collection of states.

Discuss

1. What population or random process would our slope test refer to?
2. What can we still report *without* a p-value?

Instructor (2 min): (1) A sampling-based test needs a defined random process/model; these are all US states/DC, not a random sample of regions. (2) We can still report the fitted slope, RMSE, R^2, residual plots, and limits of interpretation.

Python: Slope Inference with `ols`

Use the model fitted in the notebook

```
import statsmodels.formula.api as smf

model = smf.ols(
    "foster ~ biological", data=twins
).fit()

print(model.params["biological"])
print(model.bse["biological"])
print(model.pvalues["biological"])
print(model.conf_int())
```

What the output means

`params`: coefficient estimates.

`bse`: coefficient standard errors.

`pvalues`: two-sided tests against zero.

`conf_int()`: 95% coefficient intervals.

`smodel.resid` supplies the residual checks; `get_prediction(...).summary_frame()` supplies mean and prediction intervals.

Takeaways: Three Different Inferences

1. **Coefficient inference**: we can test or form CIs for both β_1 (slope) and β_0 (intercept). The slope is usually more scientifically relevant.

2. **Slope CI**: $b_1 \pm t^* SE(b_1)$ estimates the population slope—not the location of an individual response.

3. **Mean CI vs. prediction interval**: a mean-response CI describes the population average at x_0; a **wider prediction interval** describes one new Y.

4. **Validity**: check residual patterns, shape, sensitivity to unusual observations, and study design. R^2 and small p-values do not establish causation or guarantee individual accuracy.